

Super Daugavet Points in Projective Tensor Products

Verification, Literature Status, and Further Transfer Results

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Abstract

The supplied argument proves a correct positive subcase of the question whether super Daugavet points pass to projective tensor products. Its key reduction is that a one-dimensional L -summand in one tensor factor turns the projective tensor product into an ℓ_1 -sum. The underlying two-coordinate ℓ_1 -stability fact is already published, so the argument is best viewed as a clean tensor corollary rather than an independent resolution.

I did not locate a proof or a counterexample to the general question in the literature available through 21 June 2026. The report records one later published enlargement and proves three further positive statements. A 2026 characterization of super Daugavet points in vector-valued L_1 spaces already implies that, for every $g \in S_{L_1(\mu)}$, the tensor $x \otimes g$ is super Daugavet whenever x is. We give a self-contained arbitrary- ℓ_1 -sum stability theorem, prove tensor transfer over both real and complex scalars for the stronger class of *ccs Daugavet points*, and prove the general super-Daugavet tensor conclusion locally at every elementary target tensor. The last result isolates the remaining obstruction: a weakly open subset of a projective tensor ball may contain no elementary tensor at all.

1 The question and the outcome

Let X be a Banach space and $x \in S_X$. The point x is a *super Daugavet point* if

$$\sup_{u \in W} \|x - u\| = 2$$

for every nonempty relatively weakly open subset W of B_X . Haller, Langemets, Perreau and Veeorg asked whether

$$x \in S_X, y \in S_Y \text{ super Daugavet} \implies x \otimes y \in S_{X \widehat{\otimes}_\pi Y} \text{ super Daugavet.} \quad (1)$$

They had already proved that a super Δ -point tensors with an arbitrary unit vector to give a super Δ -point, while Dantas, Jung and Rueda Zoca had proved the corresponding elementary-tensor transfer for ordinary Daugavet points.

Then

$$T(x_\alpha \otimes y) = \langle Ty, x_\alpha \rangle \rightarrow \langle Ty, x \rangle = T(x \otimes y),$$

because x_α converges to x weakly, and the conclusion follows. $\square$

We do not know whether super Daugavet points pass to the projective tensor product similarly to Daugavet points. However, let us point out that there is an analogue to Lemma 3.12 for those points.

Source crop from Haller–Langemets–Perreau–Veeorg, recording the open super-Daugavet tensor-transfer sentence.

Outcome of the present investigation.

- (i) The supplied theorem and proof are correct.
- (ii) The main ingredient—stability of super Daugavet points under ℓ_1 -sums—was already known in two-coordinate form. Thus the supplied tensor statement is essentially an immediate application of known results and the standard tensor identity.
- (iii) No full proof or counterexample to (1) was found in the literature searched through 21 June 2026.
- (iv) The positive results proved below substantially enlarge the supplied subcase, but they do not settle (1) in full.

All results below are valid over both the real and complex scalar fields. In the complex case, slices are defined using real parts of functionals.

2 A net criterion

We use repeatedly the following standard characterization.

Lemma 2.1 (Net criterion). *Let $x \in S_X$. The following are equivalent.*

- (a) *x is a super Daugavet point.*
- (b) *For every $u \in B_X$ there is a net $(u_\alpha) \subset B_X$ such that*

$$u_\alpha \xrightarrow{w} u \quad \text{and} \quad \|x - u_\alpha\| \longrightarrow 2.$$

Proof. Assume (a). Direct the pairs (U, ε) , where U is a relatively weak neighborhood of u in B_X and $\varepsilon > 0$, by refinement of U and decrease of ε . Choose $u_{U, \varepsilon} \in U$ with $\|x - u_{U, \varepsilon}\| > 2 - \varepsilon$. This is the required net.

Conversely, let W be a nonempty relatively weakly open subset of B_X and choose $u \in W$. A net as in (b) is eventually in W , and therefore $\sup_{v \in W} \|x - v\| = 2$. $\square$

The same proof gives the weak-star version in a dual space.

3 Audit of the supplied partial result

The supplied theorem is the following.

Theorem 3.1 (Supplied L -summand subcase). *Let X and Y be Banach spaces over $\mathbb{K}$. Suppose that $x_0 \in S_X$ is a super Daugavet point and that*

$$Y = \mathbb{K}y_0 \oplus_1 Z$$

isometrically for some $y_0 \in S_Y$. Then $x_0 \otimes y_0$ is a super Daugavet point of $X \widehat{\otimes}_\pi Y$.

Proof. The projective tensor product distributes isometrically over ℓ_1 -sums:

$$X \widehat{\otimes}_\pi (\mathbb{K}y_0 \oplus_1 Z) \cong (X \widehat{\otimes}_\pi \mathbb{K}y_0) \oplus_1 (X \widehat{\otimes}_\pi Z) \cong X \oplus_1 (X \widehat{\otimes}_\pi Z).$$

Under this isometry, $x_0 \otimes y_0$ corresponds to $(x_0, 0)$. It remains to verify that $(x_0, 0)$ is super Daugavet in $X \oplus_1 E$ for every Banach space E .

Fix $(u, v) \in B_{X \oplus_1 E}$, put $s = \|v\|$ and $r = 1 - s$, and note that $\|u\| \leq r$. If $r = 0$, then $u = 0$ and the constant net $(0, v)$ is at distance $1 + \|v\| = 2$ from $(x_0, 0)$. If $r > 0$, Lemma 2.1 gives a net $(p_\alpha) \subset B_X$ such that

$$p_\alpha \xrightarrow{w} u/r, \quad \|x_0 - p_\alpha\| \longrightarrow 2.$$

Set $w_\alpha = (rp_\alpha, v)$. Then $w_\alpha \in B_{X \oplus_1 E}$, $w_\alpha \rightarrow (u, v)$ weakly, and

$$\begin{aligned} \|(x_0, 0) - w_\alpha\| &= \|x_0 - rp_\alpha\| + s \\ &\geq \|x_0 - p_\alpha\| - (1 - r)\|p_\alpha\| + s \\ &\geq \|x_0 - p_\alpha\|. \end{aligned}$$

The left side is at most 2, so it converges to 2. Lemma 2.1 finishes the proof. $\square$

Audit verdict

Every mathematical step above is valid. In particular:

- the net characterization is correct;
- the scaling u/r is legitimate because $\|u\| \leq r$;
- the displayed distance estimate has the correct direction;
- the projective tensor identity with an ℓ_1 -sum is isometric; and
- the normalized indicator of a finite positive atom spans an L -summand in $L_1(\mu)$.

There is, however, a novelty qualification. Martín–Perreau–Rueda Zoca proved the required two-coordinate ℓ_1 -stability for super Daugavet points, and Lee–Roldán–Tag later restated it explicitly. Consequently, Theorem 3.1 is best regarded as a direct tensor-product corollary of a known stability theorem. The supplied proof remains useful because it is short, self-contained, and makes the tensor mechanism explicit.

4 Arbitrary ℓ_1 -sums

The two-coordinate fact extends to arbitrary index sets in exactly the form needed for tensor applications.

Theorem 4.1 (Complete ℓ_1 -sum stability). *Let $(E_i)_{i \in I}$ be Banach spaces and*

$$E = \left(\bigoplus_{i \in I} E_i \right)_1.$$

Let $z = (z_i)_{i \in I} \in S_E$. Assume that, whenever $z_i \neq 0$, the point $z_i/\|z_i\|$ is a super Daugavet point of E_i . Then z is a super Daugavet point of E .

Proof. Write

$$I_0 = \{i \in I : z_i \neq 0\}, \quad a_i = \|z_i\|, \quad x_i = z_i/a_i \quad (i \in I_0).$$

The set I_0 is countable and $\sum_{i \in I_0} a_i = 1$.

Fix $u = (u_i)_{i \in I} \in B_E$, and put

$$s = \sum_{i \notin I_0} \|u_i\|.$$

Choose numbers $c_i \geq \|u_i\|$ for $i \in I_0$ such that

$$\sum_{i \in I_0} c_i = 1 - s.$$

This is possible because $\sum_{i \in I_0} \|u_i\| \leq 1 - s$ and $I_0 \neq \emptyset$.

For every $i \in I_0$ with $c_i > 0$, use Lemma 2.1 to choose a net $(p_{i,\alpha_i}) \subset B_{E_i}$ satisfying

$$p_{i,\alpha_i} \xrightarrow{w} u_i/c_i, \quad \|x_i - p_{i,\alpha_i}\| \longrightarrow 2.$$

If $c_i = 0$, then $u_i = 0$ and we simply use the constant value 0. On the product directed set define $w_\alpha = (w_{i,\alpha})_{i \in I}$ by

$$w_{i,\alpha} = \begin{cases} c_i p_{i,\alpha_i}, & i \in I_0, \\ u_i, & i \notin I_0. \end{cases}$$

Then

$$\|w_\alpha\| \leq \sum_{i \in I_0} c_i + \sum_{i \notin I_0} \|u_i\| = 1,$$

so $w_\alpha \in B_E$.

We claim that $w_\alpha \rightarrow u$ weakly. Indeed, if $f = (f_i) \in E^* = (\oplus_i E_i^*)_\infty$, convergence holds on every finite set of coordinates. Outside a finite set $F \subset I_0$, the tail is bounded by

$$2\|f\| \sum_{i \in I_0 \setminus F} c_i,$$

which can be made arbitrarily small. This proves weak convergence.

For $i \in I_0$ we have

$$\|a_i x_i - c_i p_{i,\alpha_i}\| \geq \|x_i - p_{i,\alpha_i}\| - (1 - a_i) - (1 - c_i),$$

so

$$\liminf_\alpha \|a_i x_i - c_i p_{i,\alpha_i}\| \geq a_i + c_i.$$

Therefore, for every finite $F \subset I_0$,

$$\liminf_\alpha \|z - w_\alpha\| \geq \sum_{i \in F} (a_i + c_i) + s.$$

Letting F increase to I_0 gives

$$\liminf_\alpha \|z - w_\alpha\| \geq 1 + (1 - s) + s = 2.$$

The reverse bound is automatic in the unit ball, hence $\|z - w_\alpha\| \rightarrow 2$. Lemma 2.1 proves the assertion. $\square$

Corollary 4.2 (Bootstrapping through an ℓ_1 decomposition). *Let $Y = (\oplus_{i \in I} Y_i)_1$, let $x \in S_X$, and let $y = (y_i) \in S_Y$. Suppose that, for every i with $y_i \neq 0$,*

$$x \otimes \frac{y_i}{\|y_i\|} \text{ is super Daugavet in } X \widehat{\otimes}_\pi Y_i.$$

Then $x \otimes y$ is super Daugavet in $X \widehat{\otimes}_\pi Y$.

Proof. Use the canonical isometry

$$X \widehat{\otimes}_\pi \left(\bigoplus_{i \in I} Y_i \right)_1 \cong \left(\bigoplus_{i \in I} (X \widehat{\otimes}_\pi Y_i) \right)_1$$

and apply Theorem 4.1. $\square$

Corollary 4.3 (The whole scalar ℓ_1 factor). *If $x \in S_X$ is a super Daugavet point, then for every set Γ and every $y \in S_{\ell_1(\Gamma)}$, the tensor $x \otimes y$ is a super Daugavet point of*

$$X \widehat{\otimes}_\pi \ell_1(\Gamma) \cong \ell_1(\Gamma, X).$$

Proof. Under the displayed isometry, $x \otimes y$ is the family $(y_\gamma x)_{\gamma \in \Gamma}$. Every nonzero normalized coordinate is a unimodular scalar multiple of x , hence is super Daugavet. Apply Theorem 4.1. $\square$

This improves the supplied coordinate-vector corollary from e_γ to *every* unit vector of $\ell_1(\Gamma)$.

5 The whole scalar L_1 factor: a later published answer

The atomic corollary in the supplied packet was substantially superseded by a later characterization. Langemets, Lõo, Martín, Perreau and Rueda Zoca proved that $f \in S_{L_1(\mu, X)}$ is a super Daugavet point if and only if, on every atom, either f vanishes or its normalized point value is a super Daugavet point of X almost everywhere. Their result immediately gives the following complete answer for a scalar L_1 tensor factor.

Theorem 5.1 (Arbitrary scalar L_1 vector; published consequence). *Let (Ω, Σ, μ) be a measure space, let X be a Banach space, let $x \in S_X$ be a super Daugavet point, and let $g \in S_{L_1(\mu)}$. Then*

$$x \otimes g$$

is a super Daugavet point of

$$X \widehat{\otimes}_\pi L_1(\mu) \cong L_1(\mu, X).$$

Equivalently, the function $t \mapsto g(t)x$ is a super Daugavet point of $L_1(\mu, X)$.

Proof. Under the canonical isometry, $x \otimes g$ corresponds to $f(t) = g(t)x$. On every atom, either $g = 0$ almost everywhere or

$$\frac{f(t)}{\|f(t)\|} = \frac{g(t)}{|g(t)|} x$$

almost everywhere. Unimodular scalar multiples of a super Daugavet point are again super Daugavet. The cited characterization of super Daugavet points in $L_1(\mu, X)$ therefore applies. $\square$

Corollary 5.2 (An L_1 -summand in the second factor). *Suppose $Y = L_1(\mu) \oplus_1 Z$ isometrically. If $x \in S_X$ is super Daugavet, then $x \otimes (g, 0)$ is super Daugavet in $X \widehat{\otimes}_\pi Y$ for every $g \in S_{L_1(\mu)}$.*

Proof. Combine Theorem 5.1 with the isometry

$$X \widehat{\otimes}_\pi Y \cong (X \widehat{\otimes}_\pi L_1(\mu)) \oplus_1 (X \widehat{\otimes}_\pi Z)$$

and the two-coordinate ℓ_1 stability from Theorem 3.1. $\square$

Remark 5.3. *Thus the supplied normalized-atom corollary is correct, but it is far from the largest currently known L_1 statement: every unit vector in the scalar L_1 factor works. This theorem is not claimed as new here; it is a direct corollary of Proposition 5.8 in the 2026 paper cited above.*

6 A full positive theorem for ccs Daugavet points

A *convex combination of slices* (abbreviated ccs) of B_X is a set $C = \sum_{i=1}^n \lambda_i S_i$, where each S_i is a slice of B_X , $\lambda_i > 0$, and $\sum_i \lambda_i = 1$. A point $x \in S_X$ is a *ccs Daugavet point* if

$$\sup_{u \in C} \|x - u\| = 2$$

for every ccs C of B_X . Bourgain's lemma implies that every ccs Daugavet point is super Daugavet. The converse is false in general.

The stronger hypothesis supplies exactly the simultaneous coherence missing from the general tensor problem.

Theorem 6.1 (ccs tensor transfer). *Let X and Y be Banach spaces over $\mathbb{K}$. If $x \in S_X$ and $y \in S_Y$ are ccs Daugavet points, then $x \otimes y$ is a ccs Daugavet point of $X \widehat{\otimes}_\pi Y$. In particular, $x \otimes y$ is a super Daugavet point.*

Proof. Put $E = X \widehat{\otimes}_\pi Y$. Let

$$C = \sum_{i=1}^n \lambda_i S_i$$

be a ccs of B_E , where

$$S_i = \{z \in B_E : \operatorname{Re} B_i(z) > 1 - \alpha_i\}, \quad B_i \in S_{E^*}, \quad 0 < \alpha_i < 1.$$

We regard B_i as a norm-one bilinear form on $X \times Y$. Fix $\eta > 0$.

For every i , choose $a_i \in B_X$ and $b_i \in B_Y$ with $\operatorname{Re} B_i(a_i, b_i) > 1 - \alpha_i/4$, and define the slice

$$A_i = \left\{ u \in B_X : \operatorname{Re} B_i(u, b_i) > \sup_{r \in B_X} \operatorname{Re} B_i(r, b_i) - \frac{\alpha_i}{4} \right\}.$$

Every $u \in A_i$ satisfies $\operatorname{Re} B_i(u, b_i) > 1 - \alpha_i/2$. Since $-x$ is also a ccs Daugavet point, choose $u_i \in A_i$ such that, for $U = \sum_i \lambda_i u_i$,

$$\|x + U\| > 2 - \eta.$$

Choose $x^* \in S_{X^*}$, with its phase normalized, so that $\operatorname{Re} x^*(x + U) > 2 - \eta$. Then

$$\operatorname{Re} x^*(x) > 1 - \eta, \quad A := \sum_i \lambda_i (1 - \operatorname{Re} x^*(u_i)) = 1 - \operatorname{Re} x^*(U) < \eta.$$

For every i , define the slice

$$D_i = \left\{ v \in B_Y : \operatorname{Re} B_i(u_i, v) > \sup_{s \in B_Y} \operatorname{Re} B_i(u_i, s) - \frac{\alpha_i}{2} \right\}.$$

Every $v \in D_i$ satisfies $\operatorname{Re} B_i(u_i, v) > 1 - \alpha_i$. Use the ccs Daugavet property of y to choose $v_i \in D_i$ such that, for $V = \sum_i \lambda_i v_i$,

$$\|y - V\| > 2 - \eta.$$

Choose $y^* \in S_{Y^*}$, again with normalized phase, so that $\operatorname{Re} y^*(y - V) > 2 - \eta$. Then

$$\operatorname{Re} y^*(y) > 1 - \eta, \quad D := \sum_i \lambda_i (1 + \operatorname{Re} y^*(v_i)) = 1 + \operatorname{Re} y^*(V) < \eta.$$

Set $w = \sum_i \lambda_i u_i \otimes v_i$. Since $u_i \otimes v_i \in S_i$, we have $w \in C$. Put $p_i = x^*(u_i)$ and $q_i = y^*(v_i)$. Since $|p_i|, |q_i| \leq 1$,

$$\sum_i \lambda_i |p_i - 1|^2 \leq 2A, \quad \sum_i \lambda_i |q_i + 1|^2 \leq 2D.$$

Cauchy–Schwarz therefore gives

$$\begin{aligned} \left| 1 + \sum_i \lambda_i p_i q_i \right| &\leq \sum_i \lambda_i |p_i - 1| + \sum_i \lambda_i |q_i + 1| \\ &\leq \sqrt{2A} + \sqrt{2D} < 2\sqrt{2\eta}. \end{aligned}$$

Also, if $p = x^*(x)$ and $q = y^*(y)$, then $|p - 1|, |q - 1| < \sqrt{2\eta}$, and hence $\operatorname{Re}(pq) > 1 - 2\sqrt{2\eta}$.

The norm-one bilinear form $\Phi(r, s) = x^*(r)y^*(s)$ now yields

$$\begin{aligned} \|x \otimes y - w\| &\geq \operatorname{Re} \Phi(x \otimes y - w) \\ &> 2 - 4\sqrt{2\eta}. \end{aligned}$$

Letting $\eta \downarrow 0$ proves that $x \otimes y$ is ccs Daugavet. Bourgain's lemma gives the super Daugavet conclusion. $\square$

Corollary 6.2 (Two L_1 -preduals). *Let X and Y be L_1 -preduals. If $x \in S_X$ and $y \in S_Y$ are super Daugavet points, then $x \otimes y$ is a super Daugavet point of $X \widehat{\otimes}_\pi Y$.*

Proof. In an L_1 -predual, the standard diametral notions under discussion, including super Daugavet and ccs Daugavet, coincide. Apply Theorem 6.1. $\square$

Corollary 6.3. *If Banach spaces X and Y both have the Daugavet property, then every elementary tensor $x \otimes y$ with $x \in S_X$ and $y \in S_Y$ is a ccs Daugavet point of $X \widehat{\otimes}_\pi Y$.*

Remark 6.4. *The last corollary concerns elementary tensors. It does not assert that the whole projective tensor product has the Daugavet property; that broader space-level tensor question remains delicate and is known under additional hypotheses such as the weak operator Daugavet property.*

7 The elementary-target local theorem

Even without ccs hypotheses, the desired conclusion can be proved in every weak neighborhood whose center is an elementary tensor.

Theorem 7.1 (Local transfer at elementary targets). *Let X and Y be Banach spaces over $\mathbb{K}$, and let $x \in S_X$ and $y \in S_Y$ be super Daugavet points. Put $p = x \otimes y$. Let $a \in B_X$, $b \in B_Y$, and let W be a relatively weak neighborhood of $a \otimes b$ in $B_{X \widehat{\otimes}_\pi Y}$. Then, for every $\varepsilon > 0$, there are $u \in B_X$ and $v \in B_Y$ such that*

$$u \otimes v \in W \quad \text{and} \quad \|p - u \otimes v\| > 2 - \varepsilon.$$

Proof. Choose a basic relatively weak neighborhood $W_0 \subset W$ of $a \otimes b$ of the form

$$W_0 = \left\{ z \in B_{X \widehat{\otimes}_\pi Y} : |B_j(z - a \otimes b)| < \delta, \ 1 \leq j \leq m \right\},$$

where the B_j are bounded bilinear forms. After rescaling the finitely many forms, we may assume $\|B_j\| \leq 1$. Choose $\eta > 0$ so small that $2\eta < \delta$ and $4\sqrt{2\eta} < \varepsilon$.

The set

$$U = \{u \in B_X : |B_j(u - a, b)| < \eta \text{ for } 1 \leq j \leq m\}$$

is a nonempty relatively weakly open subset of B_X . Since $-x$ is super Daugavet, choose $u \in U$ with $\|x + u\| > 2 - \eta$. Now

$$V = \{v \in B_Y : |B_j(u, v - b)| < \eta \text{ for } 1 \leq j \leq m\}$$

is a nonempty relatively weakly open subset of B_Y . Since y is super Daugavet, choose $v \in V$ with $\|y - v\| > 2 - \eta$. For every j ,

$$\begin{aligned} |B_j(u, v) - B_j(a, b)| &\leq |B_j(u, v - b)| + |B_j(u - a, b)| \\ &< 2\eta < \delta, \end{aligned}$$

so $u \otimes v \in W_0 \subset W$.

Choose $x^* \in S_{X^*}$ and $y^* \in S_{Y^*}$, with phases normalized, such that

$$\operatorname{Re} x^*(x + u) > 2 - \eta, \quad \operatorname{Re} y^*(y - v) > 2 - \eta.$$

Thus each of $x^*(x)$, $x^*(u)$ and $y^*(y)$ is within $\sqrt{2\eta}$ of 1, while $y^*(v)$ is within $\sqrt{2\eta}$ of -1 . For the norm-one bilinear form $\Phi(r, s) = x^*(r)y^*(s)$, it follows that

$$\operatorname{Re} \Phi(x \otimes y - u \otimes v) > 2 - 4\sqrt{2\eta} > 2 - \varepsilon.$$

Hence $\|x \otimes y - u \otimes v\| > 2 - \varepsilon$. $\square$

Corollary 7.2. *If every nonempty relatively weakly open subset of $B_{X\widehat{\otimes}_\pi Y}$ contains an elementary tensor, then super Daugavet points $x \in S_X$ and $y \in S_Y$ produce a super Daugavet tensor $x \otimes y$.*

The hypothesis of this corollary is generally false in the strongest possible way. Petitjean proved that the set

$$\{a \otimes b : a \in X, b \in Y\}$$

is weakly closed in $X\widehat{\otimes}_\pi Y$. Thus every non-elementary tensor has a weak neighborhood containing no elementary tensor. This shows precisely why an argument based only on selecting one elementary tensor from the target weak open set cannot settle the general problem.

8 Why the remaining step is genuinely different

The known and new information can be summarized as follows.

Input	Known conclusion for $x \otimes y$
x super Δ , y arbitrary	super Δ
x, y Daugavet	Daugavet
x, y ccs Daugavet	ccs Daugavet, hence super Daugavet
x super Daugavet, $y \in L_1(\mu)$ arbitrary	super Daugavet
x, y super Daugavet, target weak neighborhood contains an elementary tensor	an elementary tensor in that neighborhood can be chosen almost antipodal to $x \otimes y$
x, y super Daugavet, arbitrary target weak neighborhood	open in general

Two tempting shortcuts fail.

The two known transfers do not combine automatically. From existing results, if x and y are super Daugavet, then $x \otimes y$ is both a Daugavet point and a super Δ -point. There is no general theorem identifying the conjunction of those two pointwise properties with the super Daugavet property: they control slices and weak neighborhoods containing the point, respectively, rather than all relatively weakly open subsets. Thus the two conclusions alone do not settle the question.

Finite convex decompositions require coherence. Every element of $B_{X\widehat{\otimes}_\pi Y}$ is weakly approximable by convex combinations of elementary tensors. Perturbing each factor separately using the super Daugavet property gives individual norming functionals, but these functionals need not be compatible across the different summands. The proof of Theorem 6.1 succeeds because the ccs hypotheses create common functionals for the two weighted families. A super Daugavet point need not be a ccs Δ -point, so this coherence cannot be assumed in the general problem.

This obstruction is consistent with recent work on weakly open sets in tensor products: slices always contain elementary tensors, while arbitrary weakly open subsets need not. That distinction is identified in the recent literature as the central difficulty in transferring diameter-two and Daugavet-type properties to projective tensor products.

9 Literature search and status as of 21 June 2026

The search included the exact sentence from the source paper, combinations of “super Daugavet”, “projective tensor product”, and “elementary tensor”, the papers citing the source that were

identifiable in bibliographic and full-text searches, the 2024 monograph on diametral notions, the 2025 article on vector-valued function spaces, the 2026 paper on the super alternative Daugavet property, a 2025 doctoral thesis on Daugavet and Δ points, the 2025 monograph manuscript on the Daugavet property, and 2024–2026 papers on weakly open sets and elementary tensors in projective tensor products.

No paper located in that search states a proof or a counterexample to (1). The most relevant later developments instead give:

- a complete characterization of super Daugavet points in $L_1(\mu, X)$, which settles every scalar L_1 tensor factor;
- positive space-level tensor results under the weak operator Daugavet property;
- a precise explanation of the weak-open-set obstruction; and
- weak closedness of the elementary tensors.

Accordingly, the responsible status statement is: *the general question appears to remain open in the literature found through 21 June 2026*. The ccs-transfer theorem and the elementary-target theorem in this report were not located in that literature; they should nevertheless be regarded as new proofs proposed here, not as peer-reviewed results.

10 Conclusion

The supplied partial result is mathematically correct, but its principal stability lemma is already known. The strongest clean package established or identified here is:

- (1) arbitrary ℓ_1 -sums preserve a point assembled from super Daugavet coordinates;
- (2) the 2026 $L_1(\mu, X)$ characterization implies that $x \otimes g$ is super Daugavet for every $g \in S_{L_1(\mu)}$ whenever x is;
- (3) two ccs Daugavet points have a ccs Daugavet elementary tensor, over both real and complex scalars; and
- (4) two super Daugavet points solve the tensor problem in every weak neighborhood centered at an elementary tensor.

The unresolved part is exactly the passage to weak neighborhoods around non-elementary tensors, where no rank-one representative is available and the separate factor perturbations need not admit a common norming bilinear form.

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